

Lecture 7: Confidence intervals

Economics 326 — Methods of Empirical Research in Economics

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Point estimation

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We use the notation $E[\cdot | \mathbf{X}] = E[\cdot | X_1, \dots, X_n]$.

- Our model:
 1. $Y_i = \beta_0 + \beta_1 X_i + U_i, \quad i = 1, \dots, n.$
 2. $E[U_i | \mathbf{X}] = 0$ for all i 's.
 3. $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i 's.
 4. $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.
 5. U 's are jointly normally distributed conditional on $\mathbf{X}$.
- The OLS estimator $\hat{\beta}_1$ is a **point estimator** of β_1 .
- For our model, conditional on $\mathbf{X}$:

$$\hat{\beta}_1 \sim N\left(\beta_1, \text{Var}\left(\hat{\beta}_1 | \mathbf{X}\right)\right),$$
$$\text{Var}\left(\hat{\beta}_1 | \mathbf{X}\right) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- With probability **one**, we have that $\hat{\beta}_1 \neq \beta_1$.

Interval estimation problem

- We want to construct an **interval estimator** for β_1 :
 - The interval estimator is called a **confidence interval** (CI).
 - A CI contains the **true** value β_1 **with some pre-specified probability** $1 - \alpha$, where α is a small probability of error.
 - For example, if $\alpha = 0.05$, then the random CI will contain β_1 with probability 0.95.
- $1 - \alpha$ is called the **coverage probability**.
- Confidence interval: $CI_{1-\alpha} = [LB_{1-\alpha}, UB_{1-\alpha}]$. The lower bound (LB) and upper bound (UB) should depend on the coverage probability $1 - \alpha$.
- The formal definition of CI: It is a **random interval** $CI_{1-\alpha}$ such that conditionally on $\mathbf{X}$,

$$P(\beta_1 \in CI_{1-\alpha} | \mathbf{X}) = 1 - \alpha.$$

Note that the random element is $CI_{1-\alpha}$.

- Sometimes, a CI is defined as $P(\beta_1 \in CI_{1-\alpha}) \geq 1 - \alpha$.

Symmetric CIs

- One approach to constructing CIs is to consider a **symmetric** interval around the estimator $\hat{\beta}_1$:

$$CI_{1-\alpha} = [\hat{\beta}_1 - c_{1-\alpha}, \hat{\beta}_1 + c_{1-\alpha}]$$

- The problem is choosing $c_{1-\alpha}$ such that $P(\beta_1 \in CI_{1-\alpha} \mid \mathbf{X}) = 1 - \alpha$.
- In choosing $c_{1-\alpha}$ we will be relying on the fact that given our assumptions and conditionally on $\mathbf{X}$:

$$\begin{aligned}\hat{\beta}_1 &\sim N(\beta_1, \text{Var}(\hat{\beta}_1 \mid \mathbf{X})), \\ \text{Var}(\hat{\beta}_1 \mid \mathbf{X}) &= \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.\end{aligned}$$

- Note that conditionally on $\mathbf{X}$:

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}} \sim N(0, 1).$$

Standard normal quantiles

- Let $Z \sim N(0, 1)$. The τ -th **quantile** (percentile) of the standard normal distribution is z_τ such that

$$P(Z \leq z_\tau) = \tau.$$

- **Median:** $\tau = 0.5$ and $z_{0.5} = 0$. ($P(Z \leq 0) = 0.5$).
- If $\tau = 0.975$ then $z_{0.975} = 1.96$. Due to symmetry, if $\tau = 0.025$ then $z_{0.025} = -1.96$.

σ^2 is known (infeasible CIs)

- **Suppose** (for a moment) that σ^2 is known, and we can compute exactly the variance of $\hat{\beta}_1$:

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Consider the following CI:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}, \right. \\ \left. \hat{\beta}_1 + z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})} \right].$$

- For example, if $1 - \alpha = 0.95 \iff \alpha = 0.05 \iff z_{1-\alpha/2} = z_{0.975} = 1.96$, and

$$CI_{0.95} = \left[\hat{\beta}_1 - 1.96 \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}, \right. \\ \left. \hat{\beta}_1 + 1.96 \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})} \right].$$

Infeasible CI validity (σ^2 known)

- We need to show that

$$P(\beta_1 \in CI_{1-\alpha} \mid \mathbf{X}) = 1 - \alpha.$$

- Next, let $\sigma_{\hat{\beta}_1} = \sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}$. Then:

$$\begin{aligned}
\hat{\beta}_1 - z_{1-\alpha/2} \sigma_{\hat{\beta}_1} &\leq \beta_1 \leq \hat{\beta}_1 + z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \\
&\Leftrightarrow -z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \leq \beta_1 - \hat{\beta}_1 \leq z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \\
&\Leftrightarrow -z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \leq \hat{\beta}_1 - \beta_1 \leq z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \\
&\Leftrightarrow -z_{1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \leq z_{1-\alpha/2}
\end{aligned}$$

Infeasible CI validity (σ^2 known)

- We have that

$$\begin{aligned}
\beta_1 &\in CI_{1-\alpha} \\
&\Leftrightarrow -z_{1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}} \leq z_{1-\alpha/2}.
\end{aligned}$$

- Let $Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 \mid \mathbf{X})}} \sim N(0, 1)$ conditionally on $\mathbf{X}$.

$$\begin{aligned}
&P(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2} \mid \mathbf{X}) \\
&= P(z_{\alpha/2} \leq Z \leq z_{1-\alpha/2} \mid \mathbf{X}) \\
&= 1 - \alpha/2 - \alpha/2 = 1 - \alpha.
\end{aligned}$$

Feasible CIs (σ^2 unknown)

- Since σ^2 is unknown, we must estimate it from the data:

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2.$$

- We can replace σ^2 by s^2 ; however, the result does not have a normal distribution anymore:

$$\begin{aligned}
\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\widehat{\text{Var}}(\hat{\beta}_1)}} &\sim t_{n-2}, \\
\text{where } \widehat{\text{Var}}(\hat{\beta}_1) &= \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.
\end{aligned}$$

Here t_{n-2} denotes the t -distribution with $n-2$ degrees of freedom.

- The degrees of freedom depend on
 - the sample size (n),
 - and the number of parameters one has to estimate to compute s^2 (two in this case, β_0 and β_1).

Feasible CIs (σ^2 unknown)

- Let $t_{df,\tau}$ be the τ -th quantile of the t -distribution with the number of degrees of freedom df : If $T \sim t_{df}$ then

$$P(T \leq t_{df,\tau}) = \tau.$$

- Similarly to the normal distribution, the t -distribution is centered at zero and is symmetric around zero:
 $t_{n-2,1-\alpha/2} = -t_{n-2,\alpha/2}.$

- We can now construct a feasible confidence interval with $1 - \alpha$ coverage as:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - t_{n-2, 1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)}, \right. \\ \left. \hat{\beta}_1 + t_{n-2, 1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)} \right],$$

where $\widehat{\text{Var}}(\hat{\beta}_1) = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$.

Example: Rent rates and average income

- Data (RENTAL.DTA): 64 cities in 1990, Rent = average rent, AvgInc = per capita income: $\text{Rent}_i = \beta_0 + \beta_1 \text{AvgInc}_i + U_i$.
- $t_{62, 0.975} \approx 2.00 \Rightarrow$ The **95%** confidence interval for β_1 is $[0.0115 - 2 \times 0.0013, 0.0115 + 2 \times 0.0013] = [0.0089, 0.0141]$.
- $t_{62, 0.95} \approx 1.671 \Rightarrow$ The **90%** confidence interval for β_1 is $[0.0115 - 1.671 \times 0.0013, 0.0115 + 1.671 \times 0.0013] = [0.0093, 0.0137]$.

The effect of estimating σ^2

- The t -distribution has heavier tails than the normal.
- $t_{df, 1-\alpha/2} > z_{1-\alpha/2}$, but as df increases $t_{df, 1-\alpha/2} \rightarrow z_{1-\alpha/2}$.
- When the sample size n is large, $t_{n-2, 1-\alpha/2}$ can be replaced with $z_{1-\alpha/2}$.

Interpretation of confidence intervals

- The confidence interval $CI_{1-\alpha}$ is a function of the **sample** $\{(Y_i, X_i) : i = 1, \dots, n\}$, and therefore is **random**. This allows us to talk about the probability of $CI_{1-\alpha}$ containing the true value of β_1 .
- Once the confidence interval is computed given the data, we have its **one realization**. The realization of $CI_{1-\alpha}$ (the computed confidence interval) is not random, and it does not make sense anymore to talk about the probability that it includes the true β_1 .
- **Once the confidence interval is computed, it either contains the true value β_1 or it does not.**